

KOPAONIK, Serbia

WMO: 133780

Lat: 43.2841N

Lon: 20.8017E

Elev: 1713

StdP: 82.37

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.0	-14.0	-20.2	0.8	-11.6	-17.4	1.0	-10.0	11.3	-3.7	10.4	-3.0	3.3	60	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	22.6	14.1	21.0	13.7	19.6	13.4	16.0	19.3	15.1	18.9	14.3	18.1	3.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	13.0	17.4	13.8	12.2	16.5	12.9	11.4	15.7	51.4	19.4	48.6	18.9	46.1	18.3	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.3	7.3	DB	-18.9	24.7	2.5	1.9	-20.7	26.0	-22.2	27.2	-23.6	28.2	-25.4	29.6	(4)
(5)			WB	-19.3	17.5	2.4	1.3	-21.0	18.4	-22.4	19.2	-23.7	19.9	-25.4	20.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-4.7	-3.9	-1.6	3.0	7.5	11.2	13.3	13.6	9.3	5.5	1.3	-3.0	(6)	
	DBStd	7.77	4.89	4.84	4.61	4.23	3.93	3.81	3.29	3.42	4.02	4.53	4.87	4.78	(7)	
	HDD10.0	2430	455	391	361	211	96	30	8	8	58	147	261	404	(8)	
	HDD18.3	5110	713	624	619	459	335	213	158	149	270	397	510	662	(9)	
	CDD10.0	369	0	0	0	2	19	68	112	121	38	8	1	0	(10)	
	CDD18.3	8	0	0	0	0	0	1	3	4	1	0	0	0	(11)	
	CDH23.3	23	0	0	0	0	0	1	8	13	1	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	3.3	3.5	3.8	3.8	3.6	3.3	2.9	2.8	2.8	3.0	3.2	3.7	3.5	(14)	
(15) Precipitation	PrecAvg	876	55	56	72	81	101	90	78	62	81	69	65	66	(15)	
	PrecMax	1262	122	91	143	203	200	188	181	150	198	164	132	122	(16)	
	PrecMin	566	16	18	17	42	44	31	15	0	18	2	2	6	(17)	
	PrecStd	161	32	21	32	35	41	38	42	44	46	36	35	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	8.7	10.8	16.0	19.4	22.6	24.5	25.2	21.6	17.6	14.9	9.7	(19)	
		MCWB	1.4	3.0	4.7	8.3	11.8	14.7	14.4	14.0	13.1	9.1	7.1	2.2	(20)	
	2%	DB	4.7	6.0	8.0	13.1	16.7	20.6	22.4	22.8	19.1	15.5	11.4	6.1	(21)	
		MCWB	0.7	1.3	2.9	7.0	10.8	14.0	14.1	14.1	12.0	9.2	5.6	2.6	(22)	
	5%	DB	2.9	4.0	6.3	10.7	15.2	19.1	20.7	21.1	17.0	13.6	9.1	4.3	(23)	
		MCWB	0.6	0.5	2.4	5.7	10.0	13.6	13.9	13.5	11.3	8.4	4.4	1.6	(24)	
	10%	DB	1.6	2.1	4.7	8.8	13.7	17.6	19.2	19.5	15.2	11.6	7.3	2.8	(25)	
		MCWB	0.2	0.2	1.8	4.8	9.6	13.0	13.5	13.2	10.6	7.9	4.3	1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	4.4	5.7	9.2	13.4	17.0	17.4	17.0	14.5	11.6	9.0	5.8	(27)	
		MCDB	4.9	7.1	8.7	13.6	16.5	20.5	20.6	19.9	19.3	14.3	12.5	6.4	(28)	
	2%	WB	2.1	2.7	4.1	7.7	12.0	15.5	16.0	15.7	13.2	10.5	7.3	3.7	(29)	
		MCDB	3.2	4.5	6.6	11.6	15.2	18.7	19.3	19.5	16.6	13.2	9.1	4.5	(30)	
	5%	WB	1.2	1.4	3.2	6.6	11.0	14.4	15.1	14.9	12.2	9.6	6.1	2.5	(31)	
		MCDB	2.2	2.9	5.3	9.8	13.9	17.8	18.8	18.9	15.4	12.0	7.5	3.5	(32)	
	10%	WB	0.2	0.5	2.1	5.5	10.0	13.4	14.2	14.1	11.4	8.7	5.0	1.5	(33)	
		MCDB	1.2	1.8	4.1	8.0	12.9	16.8	17.9	18.1	14.3	10.9	6.3	2.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.5	5.8	6.0	6.5	6.9	7.5	7.6	6.4	6.4	5.5	5.2	(35)	
		MCDBR	7.2	7.4	7.8	9.0	8.7	9.0	9.7	9.6	8.9	9.0	8.4	7.5	(36)	
	5% WB	MCWBR	4.7	4.6	4.5	4.7	5.0	5.1	5.0	4.9	4.6	5.2	5.3	4.9	(37)	
		MCWBR	5.6	5.9	6.7	8.1	7.8	8.2	8.4	8.3	7.4	7.4	5.7	5.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.266	0.286	0.316	0.357	0.353	0.346	0.346	0.354	0.338	0.311	0.285	0.265	(40)	
		taud	2.464	2.407	2.401	2.310	2.353	2.395	2.384	2.358	2.403	2.492	2.516	2.490	(41)	
	Edn at Noon		885	919	928	908	918	923	918	900	890	878	854	853	(42)	
		Edh at Noon	71	90	103	123	121	117	117	116	102	82	68	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.45	2.27	3.38	4.36	5.32	6.06	6.26	5.68	3.98	2.77	1.76	1.25	(44)	
	RadStd		0.22	0.28	0.42	0.54	0.52	0.58	0.48	0.52	0.39	0.36	0.20	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	N/A	+0.59	+0.87	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		+0.60	N/A	N/A	N/A	+0.35	N/A	N/A	-120	N/A	N/A

Nomenclature: See separate page